COMPUTATIONAL MATHEMATICS (SCSI1113)
SEMESTER x 20xx/20xx
TEST 1

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET

Question 1 **18 marks**

Given the following linear system,

$$x - y - 2z = 3$$
$$-x + 2y + 3z = 1$$
$$2x - 2y - 2z = -2$$

a) Form the coefficient matrix A for the linear system. (1 mark)
b) Find inverse of A, (A^{-1}) using cofactor matrix. Use row 1 expansion to obtain the determinant. (7 marks)
c) The general form of LU matrix is given as follow. Find the solution of the following linear equation using LU factorization in Crout form matrix. (10 marks)

Question 2 **15 marks**

a) Given
$\mathbf{u} = (-4, 0, 2, -2)$ $\mathbf{v} = (2, 0, -1, 1)$
i. Find $\|2\mathbf{u}\| - 2\|\mathbf{u}\|$ (3 marks)
ii. Find the dot product of vectors **u** and **v** (3 marks)
iii. Find 5(**v**+3**u**) -3**v** (3 marks)
b) Find the distance between vector **u** = (0,2,2) and **v**=(2,0,1) (3 marks)
c) Given **u** = (5, –1, –2) and **v** = (–2, 4, –2), compute **u** × **v** (3 marks)

Question 3 **17 marks**

Let $M_1 = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$, $M_2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$, $M_3 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

a) Show that the matrices are linearly independent. (5 marks)

b) Find the unique scalars c_1, c_2, c_3 such that the matrix $M = \begin{bmatrix} 3 & 5 \\ 4 & 3 \end{bmatrix}$ can be written as

$M = c_1M_1 + c_2M_2 + c_3M_3$. (5 marks)

c) Show that the matrix $M = \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$ cannot be written as a linear combination of M_1, M_2, and M_3. (4 marks)

d) Given $M_4 = \begin{bmatrix} 0 & 1 \\ 1 & d \end{bmatrix}$, what is the value of d so that M_2, M_3 and M_4 are linearly independent. (3 marks)